

Keshav Jindal

Backend Systems Engineer — Distributed Systems, High-Performance Java & Cloud-Native

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EDUCATION

Maharaja Agrasen Institute of Technology (GGSIPO), New Delhi

Aug 2023 – May 2027

Bachelor of Technology in Computer Science and Engineering

CGPA: 7.78

Note: At least 1 semester remaining following the completion of summer 2026 internships.

EXPERIENCE

Software Engineer Intern (Backend), Special Projects Group — Visa

Jun 2026 – Aug 2026

- Architected fault-tolerant, event-driven distributed systems using **Java 17**, **Spring Boot**, and **Apache Kafka**, scaling core payment processing pipelines to handle **25,000+ Transactions Per Second (TPS)**.
- Re-engineered legacy synchronous REST APIs to asynchronous **gRPC/Protobuf** endpoints, dropping **p99 latency by 60%** (from 120ms to 48ms) and drastically reducing network payload overhead.
- Orchestrated enterprise **AWS Kubernetes (EKS)** deployments, implementing **OpenTelemetry** and **Jaeger** for distributed tracing to resolve cross-service microservice bottlenecks with **99.99% SLA uptime**.

Backend Engineer Intern — Network People Services and Technologies

Jun 2024 – Aug 2024

- Designed robust financial microservices implementing **CQRS** (Command Query Responsibility Segregation) and **Event Sourcing** patterns, successfully ensuring strict ACID compliance for **100,000+ daily ledger updates**.
- Optimized **PostgreSQL** read-replicas by integrating **Redis** distributed caching and **PgBouncer**, cutting database I/O bottlenecks and accelerating highly concurrent query throughput by **300%**.
- Hardened system security using **Spring Security (OAuth2.0/OIDC)** and implemented **Distributed Locking (Redisson)** to prevent race conditions during parallel transaction processing.

PROJECTS

High-Frequency Trading (HFT) Order Matching Engine — *Java Concurrency, LMAX Disruptor, WebSockets*

- Engineered a low-latency limit order book utilizing **Lock-Free Data Structures**, **Atomic Variables**, and custom **G1GC/ZGC** JVM tuning to eliminate thread contention and minimize Garbage Collection pauses.
- Achieved deterministic execution with an end-to-end processing latency of **sub-5ms** for **1 Million+ mock events/sec**, streaming real-time market data directly to clients via non-blocking **WebSockets**.

Distributed E-Commerce Saga Orchestrator — *Spring WebFlux, Kafka, MongoDB, Docker*

- Built a fully reactive, non-blocking order management system using **Spring WebFlux** and **Project Reactor**, optimizing CPU thread-pool utilization under extreme simulated load.
- Implemented the **Saga Design Pattern** and **Idempotent API** design across distributed nodes using **Kafka consumer groups**, guaranteeing absolute data consistency and automated rollbacks during node failures.

ACHIEVEMENTS & COMPETITIONS

- Data Structures & Algorithms:** Solved **500+ algorithmic problems** on LeetCode, mastering advanced Graph theory, Dynamic Programming, and time-space optimization critical for system-level programming.
- System Architecture & Design:** Expert in applying **SOLID principles**, **CAP Theorem** trade-offs, and horizontal scaling strategies to architect highly available, partition-tolerant backend environments.
- DevOps & SRE Mastery:** Proactively leverage **Prometheus** and **Grafana** for real-time system monitoring, ensuring high reliability and applying **Test-Driven Development (TDD)** for 90%+ code coverage.

TECHNICAL SKILLS

Languages & Core: Java (11/17/21), C/C++, SQL, JVM Internals, Multithreading, Lock-Free Concurrency

Backend Frameworks: Spring Boot, Spring WebFlux, Spring Cloud, Hibernate, gRPC, Protobuf, Node.js

Architecture: Microservices, Event-Driven, CQRS, Saga Pattern, REST API Design, CAP Theorem, ACID

Databases & Messaging: PostgreSQL, MongoDB, Redis, Apache Kafka, RabbitMQ, Elasticsearch

Cloud & Observability: AWS (EKS, S3, RDS), Docker, Kubernetes, OpenTelemetry, Prometheus, Grafana

DevOps & Tools: Git, CI/CD (GitHub Actions), Maven, Gradle, JUnit, Mockito, Linux, Agile/Scrum